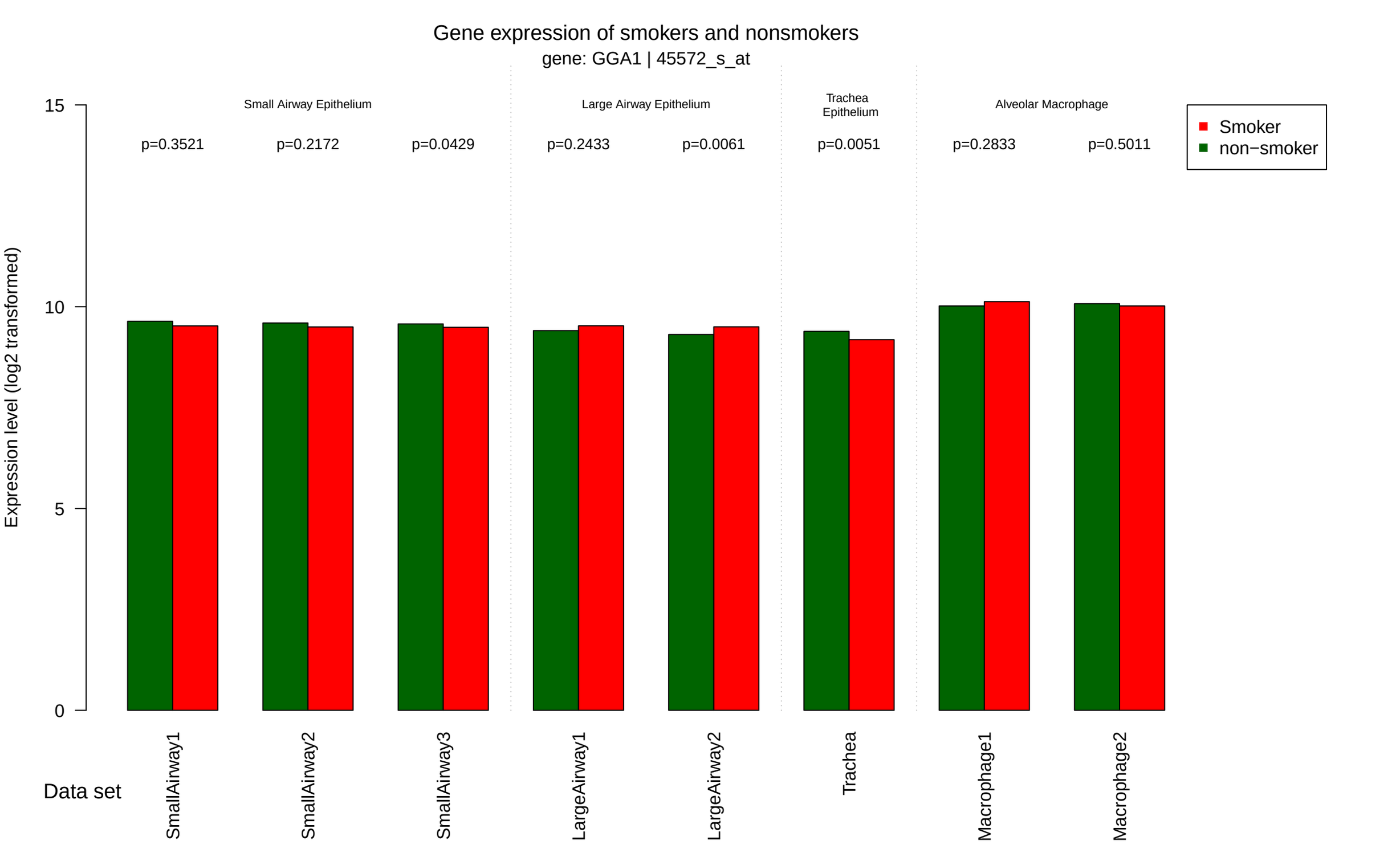
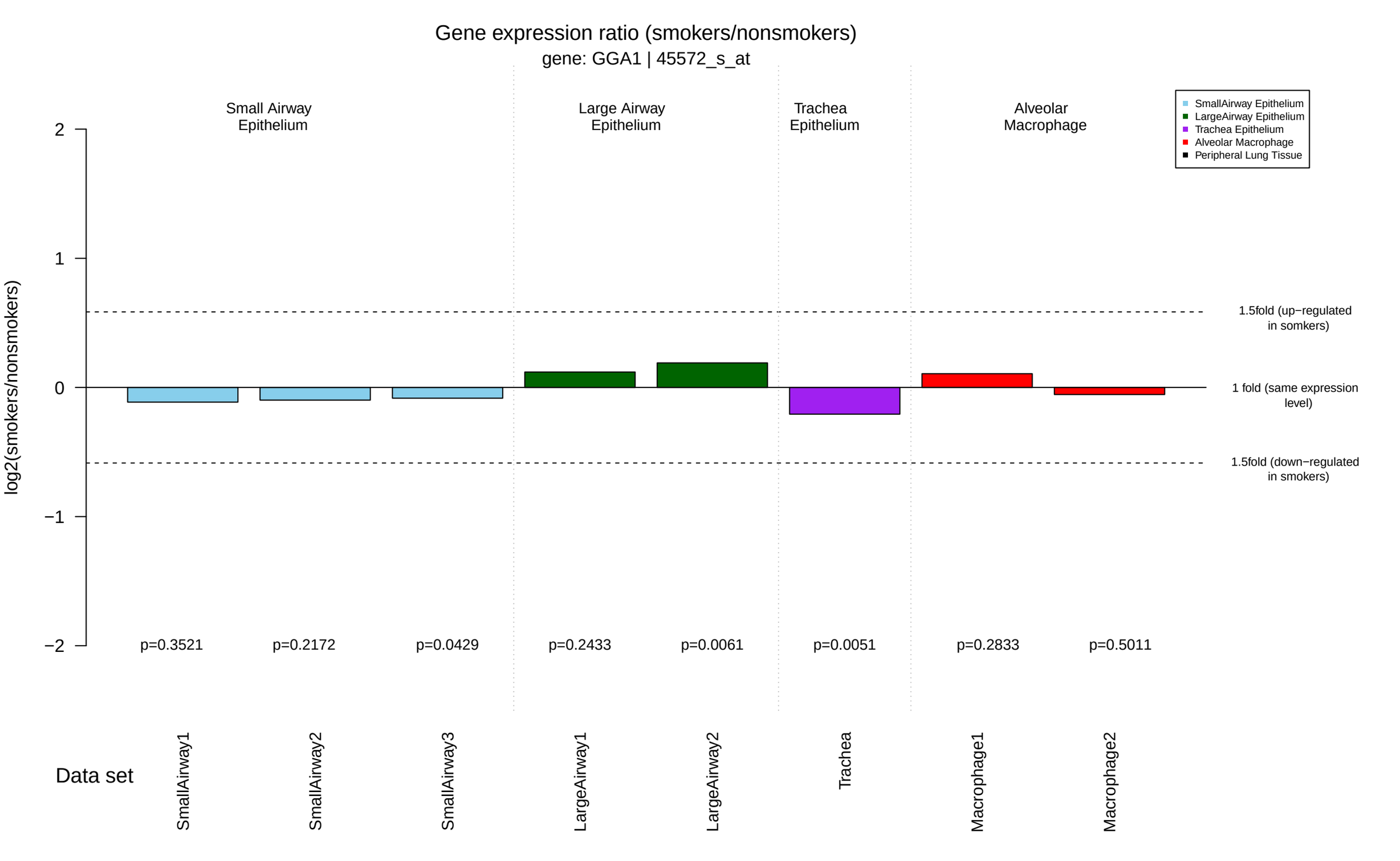
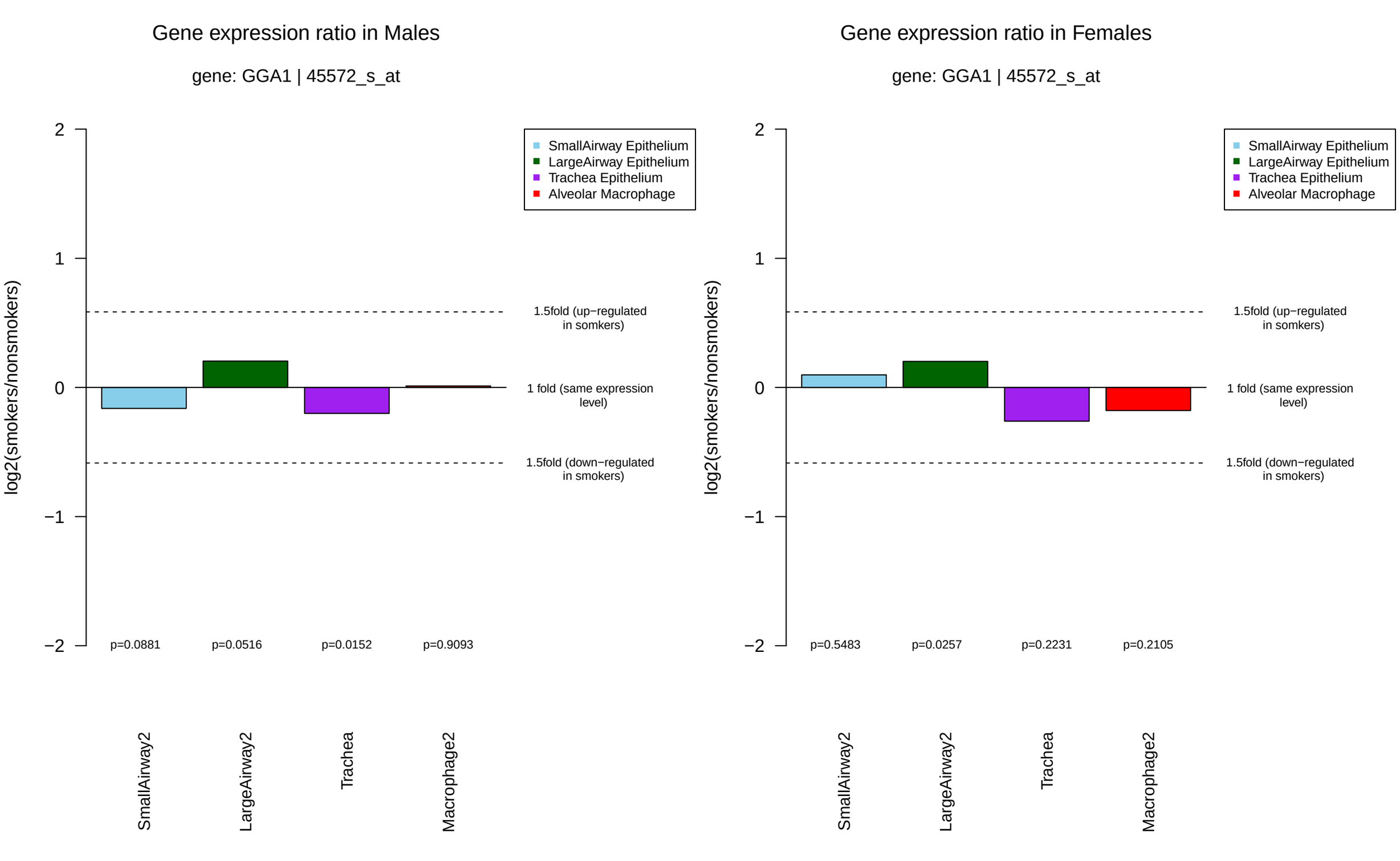
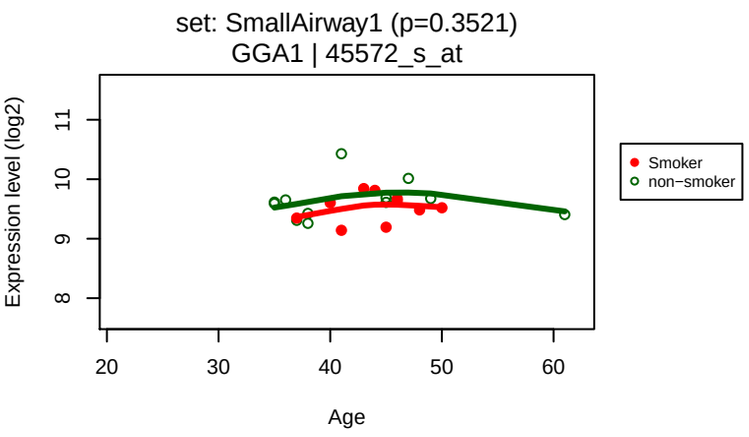
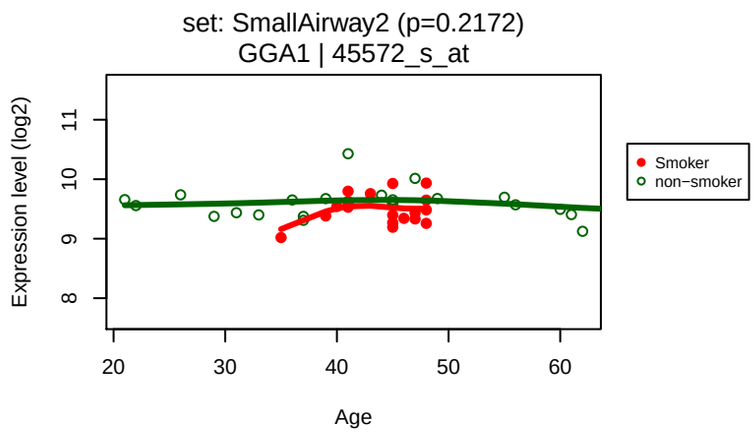
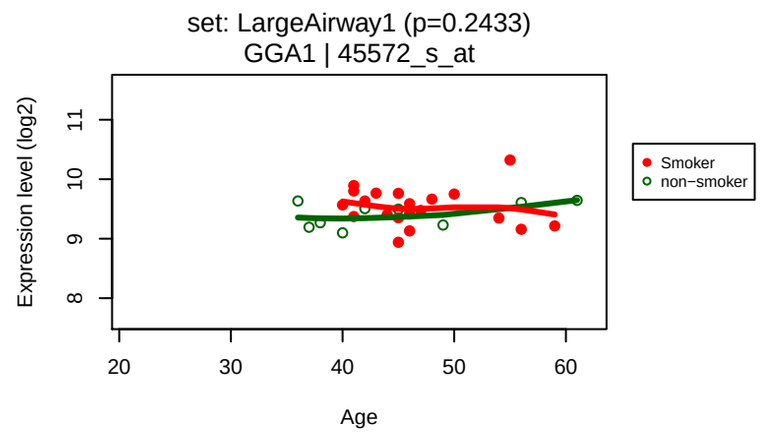

Gene expression of smokers and nonsmokers
gene: GGA1 | 45572_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3521
p=0.2172
p=0.0429
p=0.2433
p=0.0061
p=0.0051
p=0.2833
p=0.5011
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

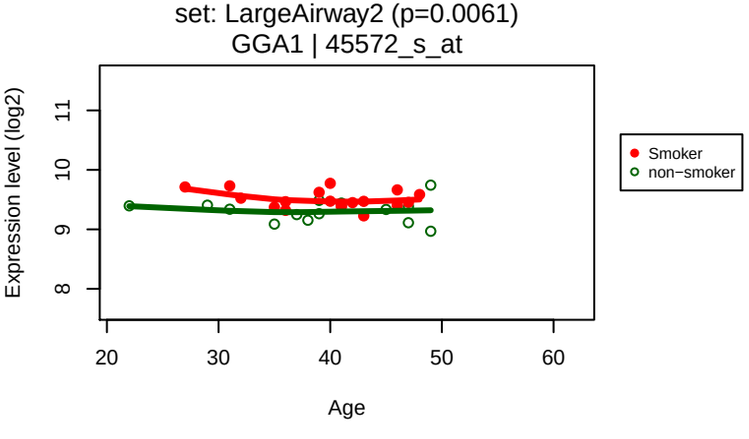

Gene expression ratio (smokers/nonsmokers)
gene: GGA1 | 45572_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3521
p=0.2172
p=0.0429
p=0.2433
p=0.0061
p=0.0051
p=0.2833
p=0.5011
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

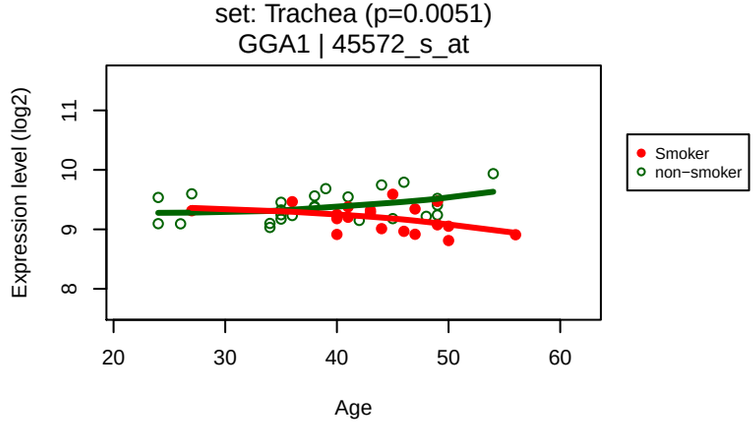

Gene expression ratio in Males
gene: GGA1 | 45572_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0881
p=0.0516
p=0.0152
p=0.9093
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA1 | 45572_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5483
p=0.0257
p=0.2231
p=0.2105
SmallAirway2
LargeAirway2
Trachea
Macrophage2

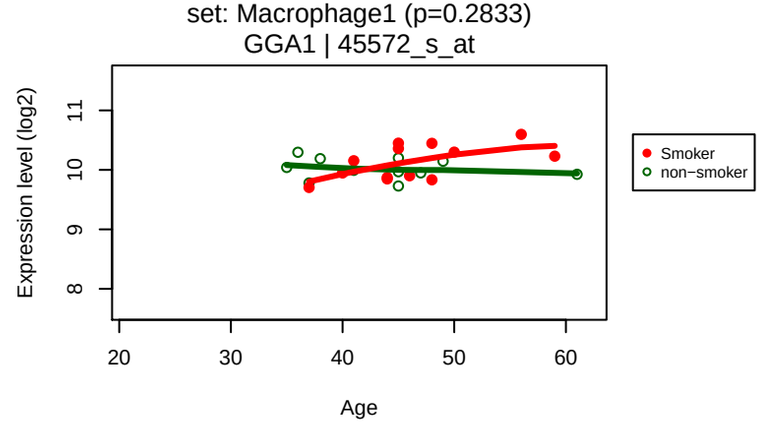

set: SmallAirway1 (p=0.3521)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker

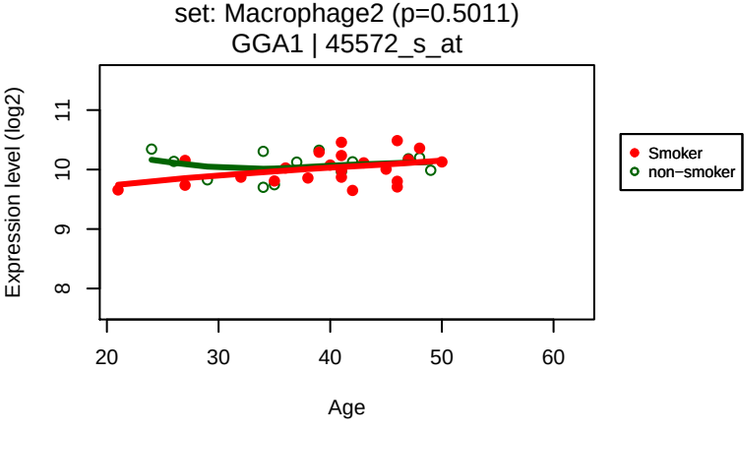

set: SmallAirway2 (p=0.2172)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker

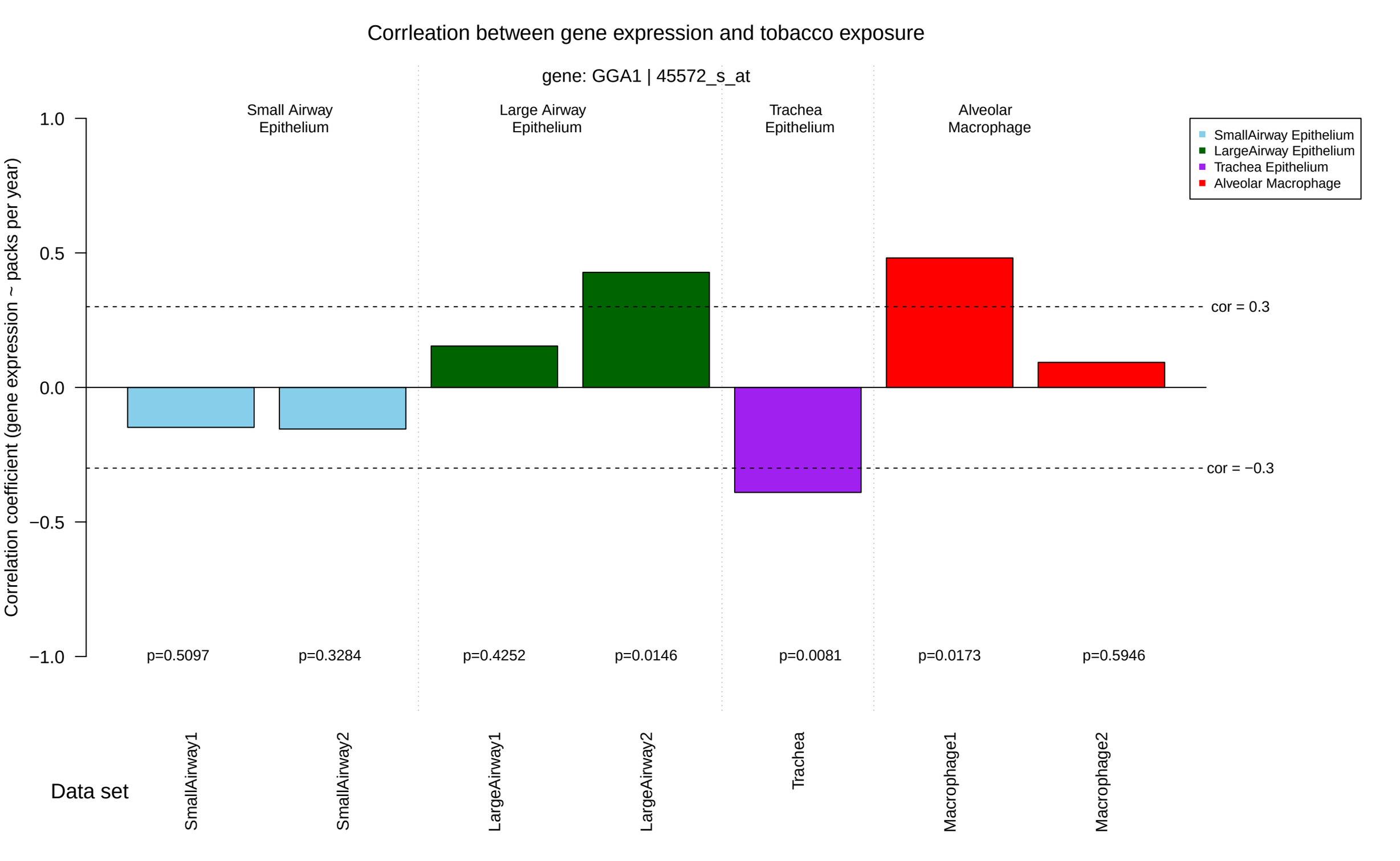

set: LargeAirway1 (p=0.2433)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0061)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2833)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5011)
GGA1 | 45572_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA1 | 45572_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5097
p=0.3284
p=0.4252
p=0.0146
p=0.0081
p=0.0173
p=0.5946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2